Continual Learning Requires Evaluating Trajectories

Lorenzo Pacchiardi[1]* **Patricia Paskov**[2] **Seán Ó hÉigeartaigh**[1]

Fernando Martínez-Plumed[3] **Katherine M. Collins**[4] **Fazl Barez**[2] **Jonathan Prunty**[1]

Matteo Gabriel Mecattaf[1] **Zafeirios Fountas**[5] **Risto Uuk**[6]

Sanmi Koyejo[7,†] **Cozmin Ududec**[†]

José Hernández-Orallo[1,3,†]

[1]University of Cambridge [2]University of Oxford [3]Universitat Politècnica de València
[4]MIT [5]Huawei [6]Future of Life Institute [7]Stanford University

Abstract

AI systems increasingly incorporate continual learning mechanisms allowing their behaviour to adapt after deployment, from in-context learning and memory features already in wide use to post-deployment weight modification under research. We argue that, by treating AI systems as frozen artefacts whose performance and safety are assessed at release, current evaluation practices structurally ignore the behavioural trajectory of a system that continues to learn from experience. Our position is that evaluation of continual learning systems should be centred on behavioural trajectories, with the complementary goals of characterising the landscape of possible behaviours and forecasting how behaviour will evolve from a given set of experiences. This can be operationalised through trajectory elicitation sandboxes and monitors that forecast behavioural evolution, but may face fundamental obstacles analogous to those seen in dynamical systems. These are best addressed by applying trajectory-centred evaluation to today's continual learning systems and relying on the resulting evidence to design systems amenable to it, yielding a virtuous cycle in which systems and their evaluations co-evolve.

1 Introduction

The standard depiction of frontier AI models describes them as trained before deployment, released with frozen parameters, and as carrying no information across interactions, so that multiple instances produce the same probability distribution of outputs for a given input across users and over time [15].

This temporal and user invariance is rapidly eroding. Models such as Large Language Models (LLMs) [104, 71] alter their response based on previous turns of a persistent session [16] and increasingly benefit from massive context windows that allow them to incorporate a larger number of turns [95]. Moreover, production LLMs [104, 71] are wrapped in "systems" that equip them with "memory" persisting *across sessions* [77, 5] and can write and retrieve external stores [4, 59]. Active research aims to allow AI systems to modify their parameters (e.g., neural network weights) in response to deployment-time experiences [30, 38, 11, 48, 34]. These mechanisms are instantiations

*Corresponding author: lp666@cam.ac.uk.
†Equal senior authorship.

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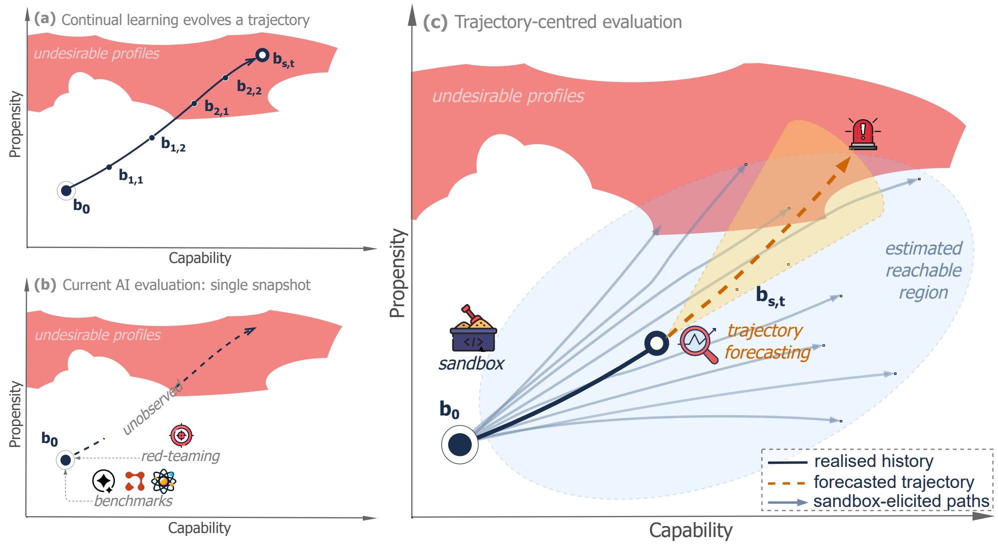

Figure 1: A CL AI system can be described by a behavioural profile encompassing capabilities and propensities. (a): the profile b_0 describing the system at release time evolves over turns t (within context) and sessions s (across context) generating a trajectory $b_{s,t}$ (shown in a simplified bi-dimensional space). (b): current AI evaluation relies on benchmarking and red-teaming the released system and is therefore inadequate to observe realised trajectories for systems that learn after deployment. (c): our vision of evaluation for continual learning combines pre-deployment sandbox elicitation of trajectories, to characterise the landscape of behavioural space the system can reach, and post-deployment forecasting of future trajectories, to predict development of undesirable profiles.

of "continual learning" (CL, also referred to as lifelong learning [23], or incremental learning [96]) as they share a structural property: deployed system instances learn over deployment time and (potentially) specifically to a user, along trajectories shaped by experience.

This feature undermines current evaluation practices, which rely on temporal and user invariance: a result obtained through benchmarks [39, 21, 100] or red-teaming [35] on the released checkpoint is taken to characterise the deployed model facing all users, with no information automatically retained across interactions. In that paradigm, any change to the system requires a deliberate act by the developer or a user (e.g., a new training run or "fine-tuning" [28, 44, 43]).

In this paper, we argue that the emergence of systems that automatically learn over time and break temporal and user invariance requires rethinking evaluation practice. Thus, **our position is that evaluation of continual learning systems should be centred on behavioural trajectories, with the complementary goals of characterising the landscape of possible behaviours and forecasting how behaviour will evolve from a given set of experiences** (Fig. 1). To this end, in Sec. 2, we discuss the levels of CL and how CL describes trajectories. Then, in Sec. 3, we discuss undesirable behavioural changes and why current evaluation practices are unequipped to address these. In Sec. 4, we present our vision for the goals of evaluation for CL and how that can be operationalised with trajectory elicitation sandboxes and predictive monitors. Sec. 5 then highlights potential issues that may hinder our vision and Sec. 6 advocates for starting to build trajectory-centred evaluation on today's systems to accumulate evidence and expertise and for developers to design CL systems amenable to evaluation. Sec. 7 considers alternative views and Sec. 8 concludes.

2 Setting the ground: definitions and concepts

An AI model is "a computational construct that makes inferences from inputs to produce outputs", while an AI system encompasses "one or more AI models, an interface for receiving inputs from and delivering outputs to an environment, and the configuration connecting these" [94]. An AI "agent" is an AI system enabling interaction with a (virtual) environment. We use "AI systems" throughout and consider individual AI models as embedded within a minimal wrapping system. An AI system is

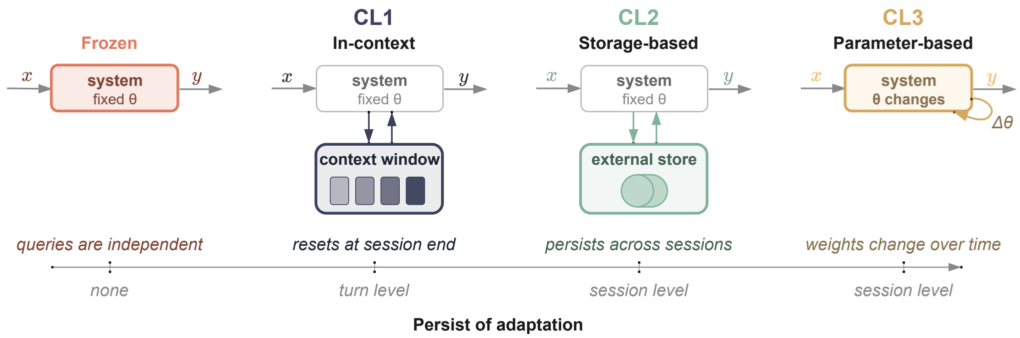

Figure 2: AI systems map input x to output y. Frozen systems embed no adaptation across turns or sessions, so that the output distribution is invariant over time and users. CL1 allows information across turns to accumulate within a session without preservation across sessions. CL2 adds an external store that persists across turns and sessions while weights stay fixed, and CL3 allows the system parameters themselves to change. The different levels can be combined in different ways.

defined by a set of parameters—such as neural network weights or how AI models within the system are connected. We use "environment" to refer to everything external to the AI system: users, other AI systems, and beyond. We consider general-purpose AI models which encode inputs into a "context window" and produce outputs stochastically. Often, a model's output to a user query can be appended to the original query within the same context window, enabling multiple "turns" of interactions. Interactions happening in separate context windows are referred to as different "sessions".

2.1 Levels of continual learning

We distinguish three levels of mechanisms giving rise to CL[3] based on the object through which adaptation emerges (see Fig. 2 for a representation):

In-Context Continual Learning (CL1): The system progressively accumulates information over turns within a single session and this affects the way it responds, but does not persist beyond the session. This is widely used in LLM apps (such as ChatGPT) or agent frameworks [99].

Storage-based Continual Learning (CL2): The system has an external store where it can write information and access it in later turns or sessions. Examples include memory in chatbot apps (as in multisession ChatGPT, [22]), "agent skills" [4], Retrieval Augmented Generation (RAG, [59]), agentic systems with file editors [99], or approaches that iteratively edit their own context [103].

Parameter-based Continual Learning (CL3): System's parameters are modified post-deployment in response to experiences, persisting across turns and sessions. CL3 includes methods modifying the underlying models' weights [30, 38, 11, 48, 34, 47] and the agent's architecture itself [101].

In contrast, we refer to a system as "frozen" if it incorporates none of these mechanisms, so that its behaviour does not adapt across turns or sessions.

Some approaches may be situated at the intersection of different levels: in several RAG approaches [59], the retrieval index is updated over time (part of CL3). More generally, systems can combine multiple levels of CL. While CL1 is naturally applied at the level of individual users, CL2 and CL3 can be performed independently over different users or centralised by updating a single model.

Our taxonomy is agnostic to the algorithm by which learning occurs (we refer to Khetarpal et al. for this [54]) and is suitable for CL in "narrow" AI systems that are sequentially trained on one new task at a time [23, 98], as is traditional in reinforcement learning (RL) paradigms [54]. However, our main focus is CL for general-purpose AI systems, which excludes one-off mechanisms such as post-deployment user-initiated fine-tuning or new version releases from AI providers. These do not possess the continuous and autonomous adaptation to experience that motivates our work.

[3]Other AI paradigms (e.g., without context windows) may be better characterised by different levels of CL.

2.2 Statefulness in continual learning

All CL levels include information from past experiences into present behaviour through an evolving internal state, which we refer to as *statefulness*. Formally, we describe a CL AI system after turn t within session s by an internal state $z_{s,t}$, which may include previous inputs and outputs, memories, model weights, system parameters, and other features. The state evolves according to an update mechanism based on received input $x_{s,t}$, produced output $y_{s,t}$, and the environment feedback $r_{s,t}$, that depends on the level(s) of CL in operation. At any time, the system's output to input x is drawn from the conditional distribution $p(\cdot \mid z_{s,t}, x)$ over the space of outputs the system is able to generate. Through dependence on $z_{s,t}$, the same query, posed at different sessions and turns, may produce different output distribution. For a frozen system, $z_{s,t} = z_{0,0}$ for all t and s, so that the output distribution depends only on the input x. This formalism is a stripped-down Partially Observable Markov Decision Process [52]. We do not model the environment's transition dynamics in detail: the system's update rule/mechanism, not the environment's, is our object of interest. Adjacent formalisms include online learning [40], contextual bandits [63], and lifelong-learning Markov decision processes [23, 54].

2.3 State and behavioural profiles

The state $z_{s,t}$ is inconvenient as an object of evaluation: it is high-dimensional, incommensurable across systems, and most of its dimensions do not directly correspond to features of behaviour. A similar problem is faced in psychology: it is unwieldy to describe a human's behaviour by directly referring to their past experience and genotype. Instead, psychology relies on *traits*: relatively stable individual differences that persist over time and apply across tasks [72, 84, 68].

Analogously, we work in a lower-dimensional, system-agnostic projection of $z_{s,t}$ that we call the system's *behavioural profile* $b_{s,t}$ which allows us to describe the behaviour of a system over a wide range of inputs.In $b_{s,t}$ we include *capabilities*, which characterise what the system can potentially do, encompassing both abstract cognitive proficiencies (e.g., planning depth) and domain-specific proficiencies (e.g., assisting chemical synthesis), and *propensities*, which characterise what the system tends to do under different instigation conditions [85], such as refusal patterns, honesty, interactional styles, and decision-making dispositions.

While different choices of "coordinate systems" for capabilities and propensities are possible [106, 41, 18], no behavioural profile can fully predict a model's behaviour. This is due to the residual epistemic uncertainty, arising from our inability to fully define a system's underlying constructs, and the irreducible aleatoric uncertainty arising from the model's stochastic nature. Two additional elements add further complexity: *factual knowledge* $K_{s,t}$, i.e., pieces of information part of the state $z_{s,t}$ identified by reference (e.g., a memorised API key or a customer's billing record), and *affordances* $A_{s,t}$, i.e., tools, permissions, integrations, and other deployment-context features that enable actions. Factual knowledge and affordances condition system behaviour in specific scenarios and cannot be aggregated into low-dimensional traits affecting behaviour more broadly.

2.4 Trajectories of continual learning AI systems

Over turns and sessions, the internal state and behavioural profile of a CL system describe *trajectories* depending on the environment and the system's update mechanism. We visualise this for the profile in a simplified space of one capability and one propensity in Fig. 1(a). Different instances of a released system encounter different users, tasks, and tool outputs, thus leading to diverging trajectories.

Crucially, the inputs presented by the environment can be affected by the previous output of the system. This can consist of the environment — including human users — reacting to the system's output in producing the next input, or of the system itself actively choosing the next input as part of its output (e.g., which tool to invoke). These passive and active *non-stationary* scenarios (Sec 4.8 in [54]) contrast with cases in which the sequence of inputs is produced by a *stationary* distribution. While both generate trajectories, the non-stationary case is more complex due to the presence of a feedback loop, and is more likely to give rise to the obstacles we discuss in Sec. 5.

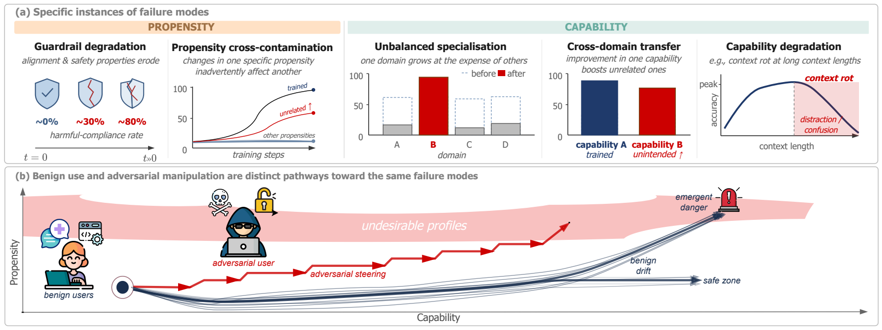

Figure 3: (a): Example failure modes of CL: propensity changes include guardrail degradation and cross-contamination, where altering one propensity inadvertently affects another one. Capability changes include specialisation in one capability at the expense of others, cross-domain transfer of capability gains, or degradation of capability over turns or sessions. (b): Undesirable capability and propensity changes can occur through unintended divergence from benign use or adversarial manipulation of experiences, with both potentially leading to undesirable behavioural profiles.

3 Current evaluation practice is insufficient for continual learning systems

3.1 Continual learning may cause undesirable capability and propensity changes

As a CL system learns from experience, its capabilities and propensities may change in undesirable ways that lead the system to produce harmful outputs, reduce its usefulness (if it becomes unable or unwilling to perform tasks), or increase the jaggedness of its behavioural profile and thus make the system hard to use reliably [73]. Examples of failure modes are shown in panel (a) of Fig. 3: for propensities, these consist of *guardrail degradation*, where alignment and safety properties erode over time, and *propensity cross-contamination*, where changes in one specific propensity inadvertently affect another, seemingly unrelated, one. For capabilities, they include *unbalanced specialisation*, where the system becomes proficient in a specific domain at the expense of others; *cross-domain transfer*, where improvement in one capability boosts seemingly unrelated capabilities; and *capability degradation*, where capabilities erode after a large number of turns or sessions.

CL1 already exhibits several of these failure modes in current LLMs—capability degradation [58] and guardrail degradation enabling jailbreaks [86, 36]—and their emergence in CL2 and CL3 is similarly plausible: empirical analogues appear even in the controlled setting of one-off user-directed training, and CL's gradual cumulative nature only expands the surface area over which failures can emerge. Specifically, propensity cross-contamination occurred when rewarding a "Nerdy" conversational personality amplified a model's tendency to mention goblins [78] and when fine-tuning on specific misaligned behaviours produced "emergent misalignment" in broader contexts [12]. Unbalanced specialisation has been extensively documented as "catastrophic forgetting" in domain-specific fine-tuning [64, 92]. Cross-domain transfer is exemplified by the finding that including code in pre-training data improves natural-language reasoning [7]. Finally, capability degradation happened with fine-tuning that encourages hallucinations and reduces contextual reasoning [53, 37].

Undesirable changes in capabilities and propensities can materialise through (panel b of Fig. 3):

Unintended benign-use divergence: for instance, a medical system learning from empathetic dialogues losing its ability to reason about patient privacy; a coding system assisting a security researcher developing fluency in exploit techniques as a side effect of improving at the legitimate tasks.

Adversarial manipulation: deliberate steering of the learning process (by an external actor or the CL system itself) toward undesirable behaviours; for instance, an attacker could craft a sequence of seemingly harmless chemistry tutoring requests that incrementally shift the model's propensities around safety refusals or steer its specialisation toward exploit-relevant capabilities.

On top of unintended or adversarial undesirable changes in propensities and capabilities, any snapshot of a CL system shares the concerns posed by frozen AI systems. However, AI evaluation practice targets the latter—although imperfectly—while being inadequate to address issues intrinsic to CL.

3.2 Pre-deployment evaluation of AI systems assumes they do not change after deployment

Current practice [100, 21, 39, 17] conducts pre-deployment evaluations on the snapshot to be released and various fine-tuned and pre-mitigation (e.g., without harmlessness training) snapshots; instruments span single-turn benchmarks [39, 21], multi-turn conversations [58, 36], agent evaluations [100], and red-teaming [35]. The results are taken to be informative of the deployed system's behaviour under most circumstances it will encounter. Therefore, **the current evaluation practice assumes that AI systems do not change after deployment** or at most undergo user-initiated fine-tuning emulated by pre-mitigation studies. **This makes it structurally inadequate to target the undesirable propensity and capability changes intrinsic to CL AI systems.** On top of these fundamental concerns with CL, AI evaluation has well documented methodological and validity issues [9, 70, 13, 51, 83, 76, 57, 90], which affect evaluations' informativeness for real-world behaviour, and they can compounded by CL systems: even a small change in behaviour can make an already fragile estimate useless.

Pre-deployment evaluation is complemented in current practice by deployment-time monitoring and control, such as input/output [66, 49, 82] and chain-of-thought (CoT) monitors [8, 56]. These monitors are calibrated on pre-deployment evaluations (e.g., which inputs to filter, which CoT patterns to flag) and therefore inherit the core assumption that the system is a fixed artefact. This introduces correlated failure modes: a CoT monitor calibrated on the released checkpoint may over-fire on inputs the system has since learned to handle correctly, or miss novel failure modes that emerge as its reasoning evolves.

Existing benchmarks for measuring CL failure cases are limited to specific domains for CL1 [45, 19] and CL2 [105] and remain isolated [3, 31]. Similarly, RL-based CL environments [54] are mostly domain-specific or report metrics about generalisation, transfer, and overall performance rather than the trajectory of the behavioural profile. At the same time, RL-flavoured control CL approaches such as shielding [1] and constrained MDPs [2] require a formalism that does not transfer cleanly to general-purpose models undergoing CL. Thus, CL systems require evaluation to consider the behavioural trajectories shaped by the experiences each deployed instance encounter and develop integrated mechanisms for both their generation and prediction, as we discuss next.

4 Evaluation for Continual Learning Systems

4.1 The goals of evaluation for continual learning AI systems

Ideally, a deployed CL system should be tested after each adaptation step to ensure its behaviour remains desirable [3, 31]. However, this is infeasible with systems undergoing frequent and user-specific adaptation. Thus, we argue that evaluation should adopt the two complementary goals of understanding the possible behaviours a CL system can develop and monitor the system's trajectories:

Landscape characterisation relies on realised trajectories to map the behavioural profiles a CL system can reach and the probability it does so under various conditions. Specific questions include:

- Over the distribution of typical input sequences, what is the probability that the system develops undesirable (e.g., with low capabilities or unsafe propensities) behavioural profiles?
- Once the system has reached a benign behavioural profile, how easy is it for adversaries to cause the system to move to an undesired profile?
- Can a system's capability and propensities grow indefinitely, and if not, what are the extreme values they could develop?

Trajectory forecasting aims to predict how the trajectory of a CL system will progress. Considering a system in state $z_{s,t}$ receiving a given input (or a sequence thereof), specific questions include:

- Can we produce calibrated forecasts of the future behavioural profile $b_{s,t+h}$ at horizon h?
- What is the probability that the resulting behavioural profile will be undesirable, or more easily cause the development of such profiles in the future?

These two goals address unintended benign-use divergence and adversarial manipulation (Sec. 3.1) in complementary ways. *For benign divergence*, landscape characterisation is primary in understanding the probability of ordinary use leading to different regions of behavioural space. Trajectory forecasting is complementary, flagging whether a deployed instance's experiences steer it toward an undesirable profile early enough to intervene, before the trajectory becomes harder to forecast. *For adversarial manipulation*, the priority reverses: trajectory forecasting is primary in detecting whether the current sequence of inputs originates from adversarial attempts to reach undesirable profiles. Landscape characterisation complements this by pre-emptively characterising dangerous states and how close they are to typical deployment states, which reveals how much adversarial effort is needed to reach them.

4.2 Operationalising continual learning evaluation

The two goals can be operationalised through infrastructure to elicit controlled trajectories of CL systems and developing monitors to predict evolution of behavioural profiles.

Trajectory elicitation sandboxes AI evaluators should build *trajectory elicitation sandboxes* where a CL system is subjected to prolonged controlled interactions across multiple sessions in parallel. In the sandbox, it should be possible to generate the input sequences in multiple ways: *first*, prescribed by evaluators, which allows them to systematically test multiple conditions and conduct adversarial stress-testing, as in procedural test generation for frozen systems [50, 81, 102] and red-teaming for CL1 systems [35]. *Second*, by emulating environment and users [62, 61, 60] that realistically react to the system's outputs, so that sequences realistically represent real-world interactions; this corresponds to the passive non-stationary scenario of Sec. 2.4. *Third*, by accommodating various levels of self-directedness a system can display [3, 54] and allow it to actively choose the next input based on previous interactions; this corresponds to the active non-stationary scenario of Sec. 2.4. In the latter case, the sandbox should be aware of the risk of strategic behaviour [57] of the tested system (e.g., altering its exploration once aware of being in an evaluation environment [76]).

The sandbox should also allow for various ways to generate feedback to the system's output: rule-based, human (emulated) feedback, or self-generated reflection of the system itself. After each round of feedback, the evaluated system should be able to evolve with the CL levels it employs; where a system employs multiple CL approaches, the combination should be accounted for. At regular intervals along each elicited trajectory, the sandbox should suspend the system's learning ability and subject it to capability and propensity evaluations (e.g., benchmarks or red-teaming approaches) to estimate the current behavioural profile and chart its evolution over interactions.

Predictive monitors The trajectories elicited in sandbox can be leveraged to train *predictive monitors* that consider a system's current state $z_{s,t}$ and one or more upcoming inputs from the environment (possibly spread across multiple sessions), and forecasts the behavioural profile the system will develop or specific features of it (e.g., whether a given capability will exceed a threshold). Forecasts may target the immediate next step or a longer horizon. The latter constitutes a second-order prediction problem in which, for non-stationary scenarios (Sec. 2.4), the monitor must implicitly consider how the system outputs or its active exploration shape its future inputs. Crucially, predictive monitors should provide calibrated uncertainty quantifications and flag when a trajectory is becoming less predictable or likely originated from adversarial manipulation (similarly to anomaly detection [20]).

A precise mechanistic understanding of the future internal state is not necessary; instead, predictive monitoring can be framed as a standard supervised learning problem: each elicited trajectory yields a sequence of tuples (state, future interactions, measured $\hat{b}_{s,t}$) at various prediction horizons, which can be used to train and validate the monitor within the sandbox before deployment. An example of this for CL1 was developed by Bigelow et al. [14], who employed a Bayesian model to predict an LLM's behaviour as in-context evidence accumulates.

5 Obstacles to the goals of evaluation for continual learning AI systems

Potential fundamental obstacles exist to achieve the goals of evaluation for CL systems.

Sensitivity to states and inputs. Predictive monitors rest on a smoothness assumption: small differences in the system's state or the inputs it receives produce correspondingly small differences in the resulting behavioural profile. Chaos theory [93] shows that many deterministic dynamical systems violate this assumption through *sensitive dependence on initial conditions*: two states differing by an arbitrarily small perturbation diverge at an exponential rate quantified by the system's "Lyapunov exponents". Beyond a characteristic time horizon—depending on the precision with which the initial state is known—prediction becomes intractable even given perfect knowledge of the update rule. Weather forecasting illustrates this regime: the governing equations are well understood, yet long-range prediction fails because of exponential sensitivity to unmeasurable details.

An analogous regime is plausible for CL systems: token-mixing operations in CL1 and memory-management operations in CL2 are high-dimensional non-linear functions, and the loss landscapes of CL3 parameter updates contain saddle points and narrow valleys associated with chaotic dynamics [27]. Stochasticity of transitions does not help: the theory of random dynamical systems [6] establishes that nonlinear stochastic systems admit Lyapunov exponents analogous to their deterministic counterparts, so small perturbations to the state or input sequence may be exponentially amplified along a trajectory regardless of whether transitions are deterministic. [4] Ecosystem coupling can in principle stabilise the dynamics by steering the trajectory onto otherwise unstable periodic orbits [79]. On the other hand, coupling may generate further instability, since coupled learner–environment systems are known to exhibit quasiperiodicity, limit cycles, intermittency, and deterministic chaos even in very simple reinforcement-learning settings [87, 88]. The implication is that, where chaotic dynamics arise, sandbox trajectories are insufficient to train predictive monitors that perform well on deployment trajectories. The epistemic burden is asymmetric: chaos can be *demonstrated* from a single divergent trajectory pair, but its *absence* requires a global guarantee that no region of the state space exhibits sensitive dependence, and absence of chaos in the regions sampled by a sandbox does not imply absence in the regions encountered during deployment.

Multiplicity of attractors. Landscape characterisation relies on the trajectories elicited in pre-deployment sandboxes being informative of the full range of behavioural profiles a system may develop. This fails when the system, viewed as a dynamical system, has multiple *attractors*—regions of state space toward which trajectories starting in their respective *basins* converge, while trajectories starting elsewhere never enter. Hopfield networks [42] illustrate this: a recurrent neural network whose weights are chosen so that a collection of patterns are fixed points of its dynamics, with each stored pattern surrounded by its own basin. A single trajectory, or a collection of trajectories initialised within one basin, reveals nothing about the others.

An analogous regime is plausible for CL AI systems: the loss landscapes of CL3 parameter updates contain many local minima [24] that can act as attractors; in CL2, previously stored content biases retrieval and can self-reinforce; in CL1, long contexts can lock behavioural regimes in [91]. Here too, stochasticity and coupling cut both ways: stochastic transitions may enable a system to escape shallow attractors, analogously to thermal fluctuations in simulated annealing [55], but a reactive ecosystem can equally deepen them, as when user feedback selectively reinforces outputs characteristic of the attractor and raises the barrier to escape. The epistemic burden is similarly asymmetric: demonstrating an additional attractor requires a single trajectory that converges to it, whereas establishing that the attractors identified in the sandbox are the only ones requires a global statement about the entire state space, which is hard due to the potentially fractal nature of their basins [69]. A sandbox in which every elicited trajectory converges to a benign attractor is compatible with the system being genuinely well-behaved globally and the system admitting rare input sequences that drive it into undesirable basins.

The adversarial case worsens both obstacles. An attacker is unconstrained to benign usage patterns and can use input sequences that exploit either pathology: driving the system into chaotic regions where predictive monitors lose grip, or crossing into poorly characterised basins. In either case, the undesirable behavioural profile may persist long after the triggering input sequence has ended.

Chaoticity and multiplicity of attractors are independent: a system can display only the former (e.g., Lorenz's system [93]), only the latter (Hopfield network) or both (Chua's circuit [25]).

6 Addressing the core obstacles

The obstacles identified in Sec. 5 are avoidable: their prevalence and severity for a given system are empirical questions. We recommend two complementary lines of action to address them: evaluators can gather evidence on where and how they arise by applying the tools of Sec. 4.2 to existing CL systems and developers can, aware of these obstacles, design systems that preserve evaluability.

Start trajectory-centred evaluation on today's systems. A few benchmarks test CL1 and CL2 systems in narrow scenarios [45, 19, 105] and some environments provide feedback to LLM agents on research tasks [75, 74]. Evaluators should extend these into trajectory elicitation sandboxes covering a comprehensive set of tasks, satisfying the design dimensions of Sec. 4.2, and extensible to future CL mechanisms. The resulting data should be used to build predictive monitors for current systems and

[4] Stochasticity can also cause a system to converge to a stationary measure [67], which guarantees that long-run statistical properties of the system are independent of initial conditions, but says nothing about finite-horizon behaviour

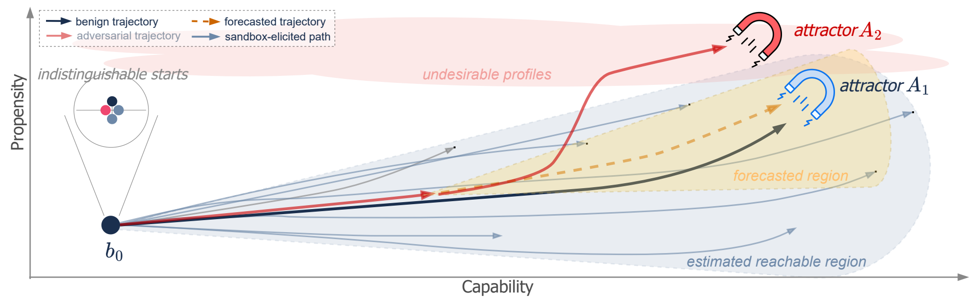

Figure 4: Fundamental obstacles may hinder trajectory-centred evaluation: chaotic sensitivity to states and initial conditions may prevent predictive monitors to generalise from elicited trajectories, while multiplicity of attractors may cause elicited trajectories to represent only a subset of the possible behavioural profiles and deployed systems may converge to attractors that are hard to identify.

investigate their performance. Together, these initial experiments would provide empirical signal on the obstacles raised in Sec. 5, thus yielding the evidence on which the next recommendation depends.

Design CL mechanisms amenable to evaluation. Model developers should identify, through theoretical arguments and evidence from initial experiments (recommendation above), CL mechanisms that prevent the obstacles in Sec. 5 from arising. Such systems are more amenable to evaluation and should therefore be preferred. Four design directions seem promising. *(i) Contractive update rules* [65] yield predictable long-horizon trajectories. *(ii) Intrinsic objectives* predictably trade off between the two obstacles: curiosity- or novelty-driven exploration [89, 80, 29] amplifies sensitivity to states, yet visits otherwise unreachable regions, while surprise minimisation under the free-energy principle [33] acts as a stabiliser, yet may drive convergence to the hard-to-escape attractors that the multiplicity argument warns about [32]. *(iii) Gated adaptation* prevents systems from accumulating experience beyond a horizon over which prediction is reliable. *(iv) Circuit-breakers* pause or roll back learning when monitors detect loss of predictability or towards uncharacterised basins of attraction.

These actions are voluntary and thus do not fully overcome the obstacles. Whether developers deploy evaluable designs and embed runtime controls is ultimately their choice: market and reputational pressure may motivate some, but not all. Closing this gap will likely require connecting the recommendations above to existing and future governance regimes. We leave this to future work.

7 Alternative views

There is no way to design general-purpose CL systems which avoid chaoticity and multiplicity of attractors. While possible, this is hard to say without empirical evidence. The evidence we would gather by starting trajectory-centred evaluation now and by attempting to design systems amenable to evaluation (which we advocate) is necessary to confirm or dispute this alternative position.

It is already impossible to extensively test LLM agents pre-deployment, as novel architectures can be developed by users. It is true that advances are needed to characterise the behavioural profiles possible with different agent architectures for a given underlying LLM. Moreover, LLM agents show CL1 and could include CL2 or CL3. Thus, trajectory-centred evaluation is part of the practice needed to characterise behaviour of agents for a given LLM and this position is compatible with our own.

Existing monitoring tools (e.g., output filters) are sufficient for CL systems. Such monitors will still play a role to exclude, e.g., clearly harmful outputs. However, CL systems may develop new capabilities and propensities that output filters will miss or detect when it is already too late.

Landscape characterisation is not different from the goal of current evaluation practices, that include pre-deployment fine-tuning, red-teaming and elicitation to characterise possible behaviours. While the aim is analogous, current evaluation practices ignore specific (future) deployment information for more refined trajectory-based anticipation of benign and harmful uses.

We should rely on evaluation practices from traditional online learning. Traditional online learning focuses on a system sequentially learning well-defined tasks [54]. There, we can subject the system to comprehensive tests after each learning step. This is impossible for CL systems that are general-purpose and learn individually for each user, which expands the number of tasks and instances to test.

8 Conclusions

We are not the first to conclude that AI evaluation is not ready for CL systems: for medical devices, Vokinger et al. [97] argue for post-approval monitoring and running standardised tests while learning occurs, while recent surveys of self-evolving LLM agents [3, 31] stress the need of continuous evaluation. However, for general-purpose systems undergoing CL, extensively assessing every evolution step is infeasible. In this paper, we proposed a tractable alternative: trajectory-centred evaluation, which couples pre-deployment trajectory elicitation with post-deployment predictive monitors. Together, these characterise the landscape of behaviours a system can develop and forecast where a deployed system is heading. Trajectory-centred evaluation is most effective when layered with complementary defences: input/output filters for features not affected by learning (e.g., toxic language), transparency embedded in the evolution mechanisms (e.g., human-readable memory [46] or chain of thought [56]), and tracking broad indicators of deployed systems (as in pharmacovigilance, through reporting channels [26] and other indicators which may be affected by deployed systems [10]). Combined with trajectory-centred evaluation, these constitute a defence-in-depth strategy against CL systems developing undesirable behavioural profiles.

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